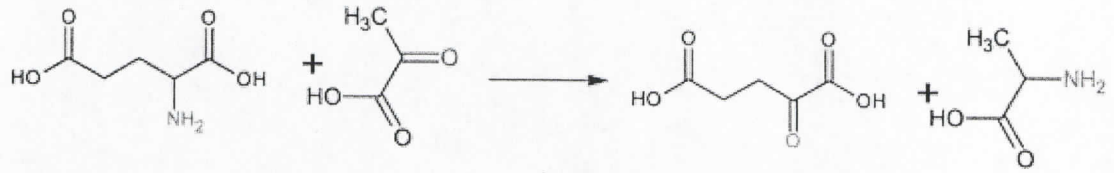
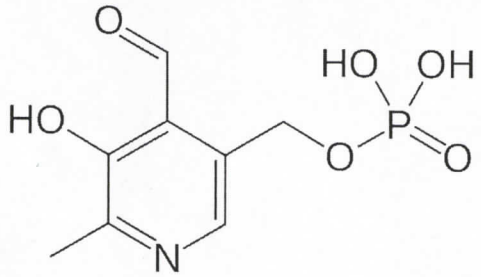
Biochemistry Cumulative Examination
October 20, 2017

1. Many of the amino acids are made from intermediates in glycolysis or the citric acid cycle. Alanine is one. The intermediate shown below is converted to alanine with the nitrogen coming from glutamate.



What is the name of the intermediate that is converted to alanine? What is the name of the intermediate that is created from the glutamic acid? 10 points

2. Enzymes that carry out transamination reactions usually use the cofactor depicted below. What is its name? It is a vitamin. Which one? 10 points



3. In the enzyme at the start of the catalytic cycle, the cofactor is usually bound as a Schiff base to a lysine. Draw this. Give another common name for Schiff bases. 15 points

4. This cofactor does many other kinds reactions than transamination. Name two other general classes of reaction that it carries out. 10 points

5. Draw a plausible reaction mechanism (i.e. 'push' electrons with arrows) to carry out the conversion shown in the drawing in question 1 using the cofactor given above. 55 points